

Assignment 5 Instructions

Two data set files dataset1.npy and dataset2.npy have been provided. These files contain semi-processed LiDAR data. Share.py file provides the framework to work with LiDAR data. Follow the instructions in the document to complete task 1-3. The source and the result should be committed to github to your personal folder. Try to separate the tasks with different version of the commit.

Link to discussion: https://emaintenancelab-my.sharepoint.com/:v:/p/jaya_kumari/ER4mFOUrtklDjlBY0Yn_xoBOXxXTcPwlhX0M46rVQ3VGg?e=11RI6n&isSPOFile=1

Outcome

- To learn processing of raw LiDAR data using Python
- To learn application of ML methods and their tuning as per the data set
- Documentation of the project on Github

Task 1 (3)

- Find the best value for the ground level
- One way to do it is using a histogram np.histogram
- Update the function get_ground_level() with your changes

Submission

For both the datasets

- Report the ground level in the readme file in your github project
- Add the histogram plots to your project Readme

Task 2 (4)

- Find an optimized value for eps.
- Plot the elbow and extract the optimal value from the plot
- Apply DBSCAN again with the new eps value and confirm visually that clusters are proper

Use the following links to understand DBSCAN and calculation of optimal epsilon value

<https://www.analyticsvidhya.com/blog/2020/09/how-dbscan-clustering-works/>

<https://machinelearningknowledge.ai/tutorial-for-dbscan-clustering-in-python-sklearn/>

Submission

For both the datasets

- Report the optimal value of eps in the Readme to your github project
- Add the elbow plots to your github project Readme
- Add the cluster plots to your github project Readme

Task 3 (5)

- Find the largest cluster, since that should be the catenary, beware of the noise cluster.
- Use the x,y span for the clusters to find the largest cluster

Submission

For both the datasets

- Report $\min(x)$, $\min(y)$, $\max(x)$, $\max(y)$ for the catenary cluster in the Readme of your github project
- Add the plot of the catenary cluster to the Readme